Cédric Girardin

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Professional Experience

Software Engineering Intern | *C++, Python* Mar 2026 – Aug 2026
AI Team | Vizrt Sports R&D *Zürich, Switzerland*
- Optimized graphics performance by porting shaders to branchless GPU vertex-pulling, Augmented sports footage to create custom datasets + added a development tool to streamline the deployment and loading of AI models.

Software Engineering Intern | *Typescript, WebGL2* Jul 2025 – Dez 2025
Web Render Engine Team | Esri R&D Zürich *Zürich, Switzerland*
- Integrated WebXR into the ArcGIS WebGL2 rendering engine, implementing performance profiling to ensure seamless spatial visualization with minimal architectural overhead

VR Software Engineer | *C#* Aug 2019 – Jul 2020
Computer Perception and Virtual Reality Lab | Bern University of Applied Sciences *Bern, Switzerland*
- Developed educational 3D renderer demonstrating full graphics pipeline from transformations to shading [Github]
- Designed one of the first VR apps in Unity for Meta Quest 1 to recognize ASL hand poses by matching stored gestures with real-time hand tracking. [Demo]
- Produced video on rasterization with Manim, with 100K+ views on Youtube [Video]

Education

ETH Zurich Zurich, Switzerland
Master of Science, Computer Science *Aug 2024 - Present*
- **Focus:** Visual and Interactive Computing, Machine Intelligence
- **Relevant Coursework:** Algorithms Lab, Computer Intelligence Lab, Physics-Based Simulation, Computer Graphics, Shape Modeling, Computer Vision, Probabilistic AI

ETH Zurich Zurich, Switzerland
Bachelor of Science, Computer Science *Aug 2021 – Dec 2024*
- **Thesis (Interactive Geometry Lab, Prof Sorkine-Hornung):** Integrated "Sensitive Couture" cloth simulation into Unity for Meta Quest 3, enabling real-time garment interaction in VR.
- **TA:** Introduction to Programming (Java), Led weekly exercise sessions for 20+ students.

Hong Kong University of Science and Technology Hong Kong
Visiting Student, Computer Science *Sep 2023 – Dec 2023*

Informatikmittelschule (IMS) Bern Bern, Switzerland
Certified IT Specialist (Swiss EFZ) *Aug 2017 – Jul 2020*

Projects & Leadership

WebGPU 2d-splatter | *Typescript, WebGPU* Sep 2026
- Developed a interactive 2d gaussian splatter in the web from scratch and wrote a blog about the creation process [Blog]

Projective Dynamics & Snoopy | *C++* Dec 2025
- Developed a real-time physics solver in C++ based on the Projective Dynamics paper, also adding a custom collision detection and response system for cloth simulation for the course Physics Based Simulation for CG [Video]

Reaction Diffusion Simulation | *Typescript, WebGL, GLSL* Jun 2025
- Implemented a Gray-Scott reaction-diffusion simulation from scratch using vanilla WebGL2 and GLSL fragment shaders [Link]

Design Commision President | *Computer Science Student Association ETH (VIS)* Present
- Lead a team of 10 designers in monthly design nights to create posters and merchandise

Technical Skills

Programming: C++, C#, Typescript, Python, GLSL, Java, C, PHP (Laravel), SQL, Haskell
Technologies: WebGPU, WebGL, PyTorch, TenserFlow, Unity, Git, Linux/GNU (Debian), LaTeX
Languages: English (fluent), German (native), French (native), Dutch (native), Chinese (A1)